

Websites, Portfolios, Profiles

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Summary

Aspiring mechatronics engineer with experience in robotics, automation, embedded systems, and systems integration. Skilled in Python programming, sensor integration, and multidisciplinary engineering collaboration. Committed to developing innovative solutions that inspire younger generations' interest in robotics. Actively seeking opportunities in autonomous programming, robotics, manufacturing, and self-driving vehicle technologies.

Education

Rochester Institute of Technology

Expected in June 2030

Bachelor of Science: Mechatronics Engineering Technology

Rochester, NY

- Relevant Coursework: Manufacturing Processes, Introduction to Digital and Microcontroller Systems, Machine Tools Lab, Calculus I & II, Circuits 1, Characterization of Metals Lab, Foundations of Metals
- College of Engineering Technology Honor Roll Fall 26 Recipient
- NSBE Fundraising Chair

Experience

Rochester Institute of Technology, RIT

February 2026 to Current

ITS Student Trainee

Rochester, NY

- Executed remote imaging and data backup assistance using BeyondTrust, TCTools and SentinelOne.
- Acquired proficiency in troubleshooting computer hardware and imaging software enhancing workplace efficiency by 15%.
- Strengthened problem-solving skills by engaging in hands-on challenging assignments including hardware repairs and system troubleshooting.

Council for Scientific and Industrial Research

June 2023 to July 2023

Intern

Accra, Ghana

- Utilized Python-based control systems for robotics prototypes and executed system testing to validate functionality within a 2-week timeline.
- Collaborated with a multidisciplinary engineering team to design, develop and test robotics and automation prototypes, contributing to full end-to-end delivery.
- Integrated sensors and control algorithms to ensure stable system performance.
- Trained and mentored four young robotics engineers, fostering interest in robotics and programming.

Projects

- **Project Auria** - Developed a solar-powered backpack capable of powering mobile devices. Programmed the charging module and integrated microelectronics components with system circuitry. Collaborated with multidisciplinary engineers to deliver a functioning prototype.
- **Automated Greenhouse System** - Collaborated with a multidisciplinary team to design and develop an Arduino-based autonomous monitoring greenhouse system. The system integrated 5 sensors and three actuators to enable real-time system responses. Control algorithms were implemented to maintain optimal environmental conditions through automated decision making.

Skills

- AutoCAD
- Excel proficiency
- MATLAB proficiency
- Python
- Object-Orientated Programming
- Robotics programming
- Software Development
- High-Level Programming
- Team collaboration
- Microcontroller experience (Arduino)
- Sensor Fusion
- Machine and Metal Lab Proficiency